

Aligned Mathematical Structures

Source-backed grouped formula sample

1 Expressions

Expression 1. The following expression is taken from a source-backed formula corpus.

$$F_{p;r}^{(L)}(q) = q^{-r(r-1)} \times \begin{cases} F_{p-2}^{(L)} \begin{bmatrix} \mathbf{Q}_{r,1} \\ \mathbf{0} \end{bmatrix} (\mathbf{e}_r|q) & \text{if } L \not\equiv r-1 \pmod{2}, \\ F_{p-2}^{(L)} \begin{bmatrix} \mathbf{Q}_{p-r,p} \\ \mathbf{0} \end{bmatrix} (\mathbf{e}_{p-r}|q) & \text{if } L \equiv r-1 \pmod{2}, \end{cases}$$

Expression 2. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} u(\tau) &= e^{\sqrt{\gamma^2+m^2/2}\tau} \left[u_0 \cosh(\gamma\tau) - \left(u_0 \sqrt{\gamma^2+m^2/2} - \dot{u}_0 \right) \frac{1}{\gamma} \sinh(\gamma\tau) \right] \\ v(\tau) &= e^{-\sqrt{\gamma^2+m^2/2}\tau} \left[v_0 \cosh(\gamma\tau) + \left(v_0 \sqrt{\gamma^2+m^2/2} + \dot{v}_0 \right) \frac{1}{\gamma} \sinh(\gamma\tau) \right], \end{aligned}$$

Expression 3. The following expression is taken from a source-backed formula corpus.

$$\begin{aligned} \mathbf{E}(\mathbf{x}, t) &= \frac{1}{4\pi} \int \rho_e(\mathbf{x}', t-r) \frac{\mathbf{r}}{r^3} d^3x' \\ &+ \frac{1}{4\pi} \int \mathbf{J}_m(\mathbf{x}', t-r) \times \frac{\mathbf{r}}{r^3} d^3x' - \frac{1}{4\pi} \int \frac{1}{r} \frac{\partial \mathbf{J}_m(\mathbf{x}', t-r)}{\partial t} d^3x', \end{aligned}$$

Expression 4. The following expression is taken from a source-backed formula corpus.

$$\Gamma^5 = \begin{pmatrix} 0 & 1_2 \\ 1_2 & 0 \end{pmatrix}; \quad \Gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix}; \quad \Gamma^0 = \begin{pmatrix} 1_2 & 0 \\ 0 & -1_2 \end{pmatrix}; \quad \Gamma^4 = -i\Gamma^5 = \begin{pmatrix} 0 & i_2 \\ i_2 & 0 \end{pmatrix}.$$

Expression 5. The following expression is taken from a source-backed formula corpus.

$$(II) \quad \begin{pmatrix} x_1 + iy_1 \\ x_2 + iy_2 \\ x_3 + iy_3 \\ x_4 + iy_4 \end{pmatrix} \leftrightarrow \frac{1}{\sqrt{2}} \begin{pmatrix} -(y_2 + y_4) + j(y_2 - y_4) \\ (y_1 + y_3) - j(y_1 - y_3) \\ (x_2 + x_4) - j(x_2 - x_4) \\ -(x_1 + x_3) + j(x_1 - x_3) \end{pmatrix},$$